Hypoglycemia Detection from ECG Signals

Hypoglycemia detection means identifying when a person's blood glucose drops too low, typically below 70 mg/dL (3.9 mmol/L), so you can warn the person and take immediate action such as consuming fast-acting carbohydrates. Detecting hypoglycemia is crucial because low blood sugar can cause sweating, confusion, seizures, or even loss of consciousness, and early detection reduces risk, especially during sleep or in cases of hypoglycemia unawareness when symptoms are not felt. People with diabetes are particularly at risk especially those using insulin because medication can lower glucose too much, often at night or after exercise.
Clinically, hypoglycemia is measured using invasive or minimally invasive methods such as fingerstick capillary tests and continuous glucose monitoring (CGM) sensors (a small, flexible sensor filament must be inserted under the skin to measure glucose levels). To reduce reliance on invasive monitoring, current research investigates non-invasive physiological signals most notably ECG signals as potential proxies to predict or detect hypoglycemic episodes.



Figure 1- Blood Glucose States: Hypoglycemia vs. Normal

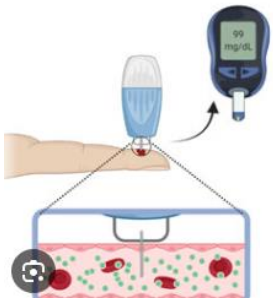


Figure 2- Finger-Stick Capillary Blood Glucose Measurement

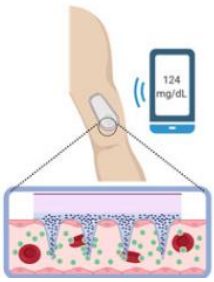


Figure 3- Continuous Glucose Monitoring (CGM) Patch

Project Overview

The goal of this project is to design, train, and evaluate a deep learning model capable of detecting hypoglycemia episodes from electrocardiogram (ECG) signals.

The dataset represents 10-second ECG windows, each containing 2,500 data points representing the electrical activity of the heart over time.
The data is collected from nine diabetic patients, each wearing both an ECG monitoring device and a Continuous Glucose Monitoring (CGM) system simultaneously.
Each record in the dataset corresponds to one ECG segment and includes the following fields:

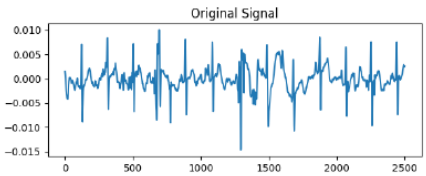


Figure 4- Raw ECG Segment (10 s, 2,500 samples)

Field Name	Description
Data points from 0 to 2499	A vector of 2,500 preprocessed and normalized ECG samples representing one 10-second segment.
label	Binary indicator of the glycemic state: 0 → Normal (Normoglycemic) 1 → Hypoglycemic
patient_id	Identifier for the subject (1–9). Used to ensure proper patient-level data splitting.
day	The day of data collection for each patient.
count	Segment index within that day.
glucose_values	The blood glucose value (mg/dL) obtained from the CGM device at the same time as the ECG window.

Before training, you must clean the dataset to remove unnecessary fields.
When splitting the data into training, validation, and testing sets, it is critical to avoid data leakage to guarantee that the model generalizes well.

Labeling Methodology
The labeling of ECG signals is derived from the Continuous Glucose Monitoring (CGM) data collected concurrently.

- The CGM system continuously measures interstitial glucose concentration at short time intervals (e.g., every 1–5 minutes).
- Each 10-second ECG window is assigned a label based on the corresponding glucose value recorded by the CGM at the same timestamp.
- If the glucose level is below the clinical threshold for hypoglycemia (typically <70 mg/dL), the window is labeled as 1 (Hypoglycemic). Otherwise, it is labeled as 0 (Normoglycemic).

This synchronization between ECG and CGM ensures that the labels accurately reflect the physiological state of the patient at the moment of ECG recording.

About ECG Signals
An Electrocardiogram (ECG) is a non-invasive recording of the heart's electrical activity over time, measured through electrodes placed on the skin. It reflects cardiac cycles, including depolarization and repolarization phases, which can be influenced by metabolic changes such as glucose level fluctuations. During hypoglycemia, the autonomic nervous system activates compensatory mechanisms that alter heart rate variability (HRV) and cardiac electrical patterns, which can be detected through subtle changes in ECG morphology and rhythm. These physiological correlations make ECG a promising biomarker for glucose monitoring.

Expected Outcome
You are expected to:

1. Design and implement a deep learning model capable of classifying each ECG segment as normoglycemic or hypoglycemic.
2. Experiment with architectures (e.g., CNN, RNN, BiLSTM, or hybrid models).
3. Select suitable training strategies (e.g., patient-wise cross-validation, class imbalance handling, or augmentation).
4. Evaluate performance using accuracy, sensitivity, specificity, and F1-score (for minority and majority classes),and AUROC, ensuring robust generalization to unseen patients.

General Guidelines

- Duration: **4 weeks**
- You are **free to choose** the architecture, loss functions, and training strategy.
- Focus on **model robustness, interpretability, and experimental justification**, not just accuracy.
- Use proper data splitting and validation practices.
- You may optionally apply data augmentation, attention mechanisms, or transfer learning if you wish.

Expected Deliverables

Deliverable	Description	Weight
Project Proposal (Nov 7)	1-page concept note outlining your approach and planned architecture	10%
Code & Notebook (Nov 28)	Fully reproducible code with documentation and comments	40%
Final Report (Nov 28)	Concise (4–6 pages): motivation, model summary, experiments, and discussion	40%
Presentation / Demo (Dec 3)	5–10-minute summary of approach and findings	10%

Evaluation Criteria

- Clarity and originality of the approach
- Soundness of experimental design
- Correctness and reproducibility of implementation
- Depth of analysis and interpretation
- Quality and clarity of report and presentation

General Notes

- You should work in groups.
- Avoid copying existing architectures without understanding or modification.
- Cite any references, papers, or code bases used.
- Questions about the correctness of your specific solution are not allowed.
- You should not bring or show your individual model to ask whether it is correct or not.
- I will continue to provide general guidance and common instructions to all of you at the beginning of each session, but NO FEEDBACK ON INDIVIDUAL SOLUTIONS BEFORE THE DEADLINE.

Note on the Use of Generative AI Tools

You are allowed to use Generative AI tools (such as ChatGPT, Grammarly, or similar) only for language correction and clarity improvement. The ideas, analysis, and implementation must be your own work.
All submitted reports will be checked using Compilatio or ithenticate (as available) to verify that no full paragraphs or sections were copied or generated wholesale.

Note on Originality and Individual Approaches

This project can be approached in hundreds of different ways, using a wide variety of deep learning architectures, preprocessing techniques, or combined hybrid models. You are encouraged to explore and experiment with your own ideas. Because of the diversity of possible methods, it is highly unlikely that two or more groups will independently develop identical or nearly identical approaches.
Therefore, if two or more submissions show very similar architectures, training strategies, or code structures, it will be assumed that ideas were shared or copied.
In such cases, I will apply a point deduction (“pointage”) for lack of originality. Please make sure to develop your own approach and keep your model design confidential until submission.